Experiment 6: Statistical Distribution Analysis of Urban Climate Data

Aim: To analyze the distributions of temperature, humidity, and precipitation using statistics and graphs.

Dataset: `urban_climate.csv`

1. Load the dataset

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("urban_climate.csv")
df.head()
```

Out[1]:

	city	country	latitude	longitude	year	month	temperature_celsius	humidity_percent
0	New York	USA	40.7128	-74.006	1980	1	6.21	48.8
1	New York	USA	40.7128	-74.006	1980	2	17.46	38.7
2	New York	USA	40.7128	-74.006	1980	3	20.28	32.3
3	New York	USA	40.7128	-74.006	1980	4	22.31	30.6
4	New York	USA	40.7128	-74.006	1980	5	20.01	22.6

2. Select suitable attributes

The selected attributes are temperature, humidity, and precipitation.

In [2]:
```
attributes = ["temperature_celsius", "humidity_percent", "precipitation_mm"]
df[attributes].describe()
```

Out[2]:

	temperature_celsius	humidity_percent	precipitation_mm
count	11040.000000	11040.000000	11040.000000
mean	13.521150	48.432944	66.811812
std	13.917589	12.792415	32.584682
min	-22.030000	20.000000	15.100000
25%	2.810000	39.400000	41.700000
50%	13.340000	48.200000	60.250000
75%	23.850000	57.500000	86.925000
max	53.650000	95.000000	179.800000

3. Calculate mean, median, and standard deviation

In [3]:
```
import pandas as pd
import numpy as np

# Select numerical columns
numeric_columns = df.select_dtypes(include=np.number).columns

# Calculate statistical measures
statistics = pd.DataFrame({
    'Mean': df[numeric_columns].mean(),
    'Median': df[numeric_columns].median(),
    'Standard Deviation': df[numeric_columns].std()
})

print("Mean, Median and Standard Deviation:")
display(statistics)
```

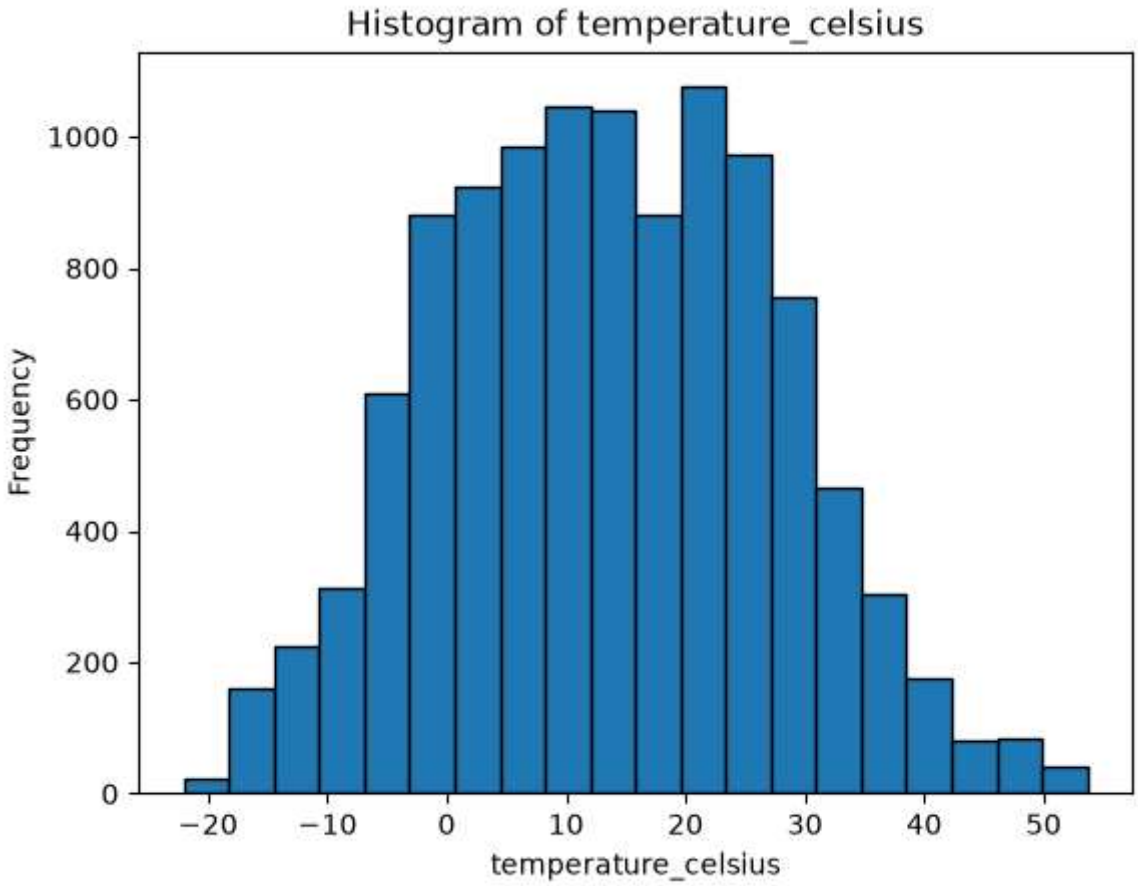
```
Mean, Median and Standard Deviation:
```

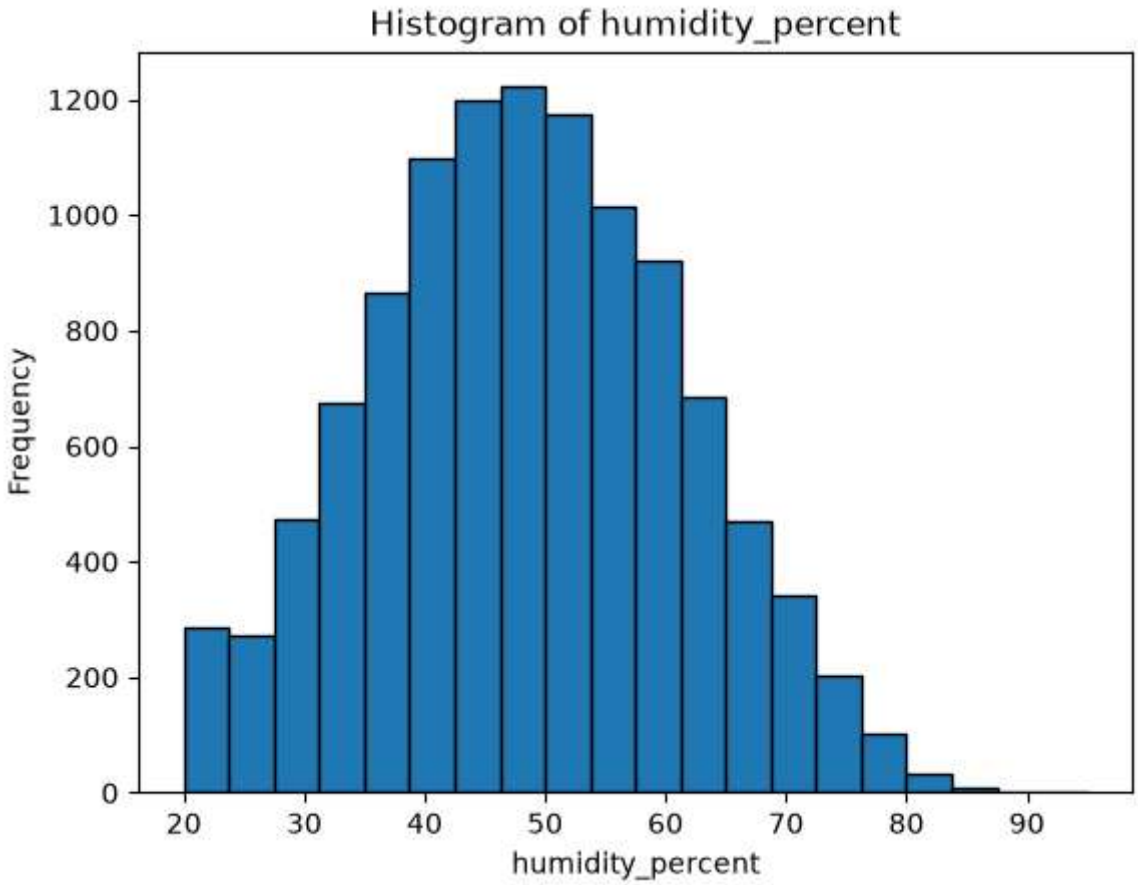
	Mean	Median	Standard Deviation
latitude	3.153551e+01	3.375030e+01	1.660261e+01
longitude	-3.587733e+01	-7.458560e+01	8.150853e+01
year	2.002500e+03	2.002500e+03	1.327652e+01
month	6.500000e+00	6.500000e+00	3.452209e+00
temperature_celsius	1.352115e+01	1.334000e+01	1.391759e+01
humidity_percent	4.843294e+01	4.820000e+01	1.279242e+01
precipitation_mm	6.681181e+01	6.025000e+01	3.258468e+01
wind_speed_ms	8.035589e+00	8.000000e+00	2.916462e+00
urban_heat_island_intensity	1.250000e+00	1.000000e+00	4.330323e-01
country_population_reference	3.963711e+08	3.192590e+08	3.911626e+08
country_area_km2	6.276432e+06	9.629091e+06	4.073504e+06
calling_code	4.130000e+01	1.700000e+01	5.582645e+01

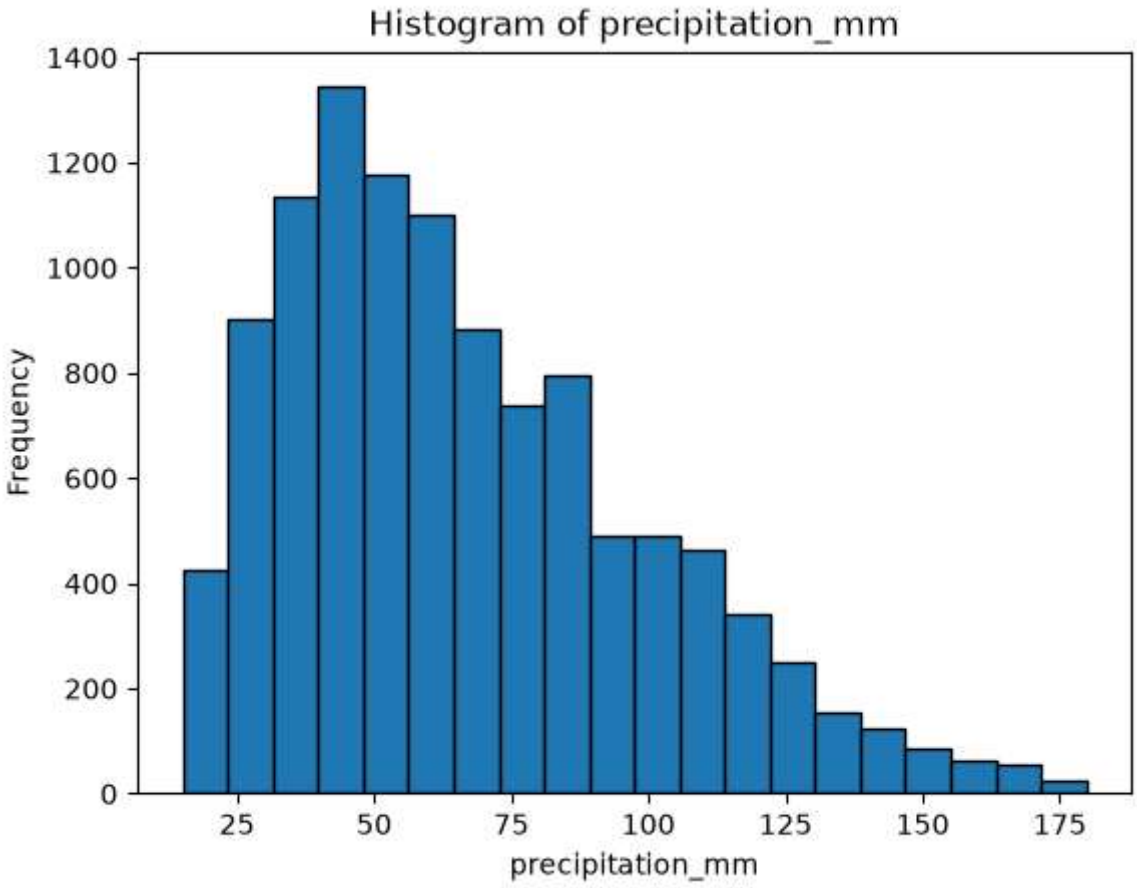
Observation: Mean and median describe the central value. Standard deviation describes how spread out the readings are. Record and compare the printed results.

4. Plot histograms

In [4]:
```
for column in attributes:
    data = df[column].dropna()
    plt.hist(data, bins=20, edgecolor="black")
    plt.xlabel(column)
    plt.ylabel("Frequency")
    plt.title("Histogram of " + column)
    plt.show()
```


Histogram of temperature_celsius
Frequency
0
200
400
600
800
1000
−20
−10
0
10
20
30
40
50
temperature_celsius

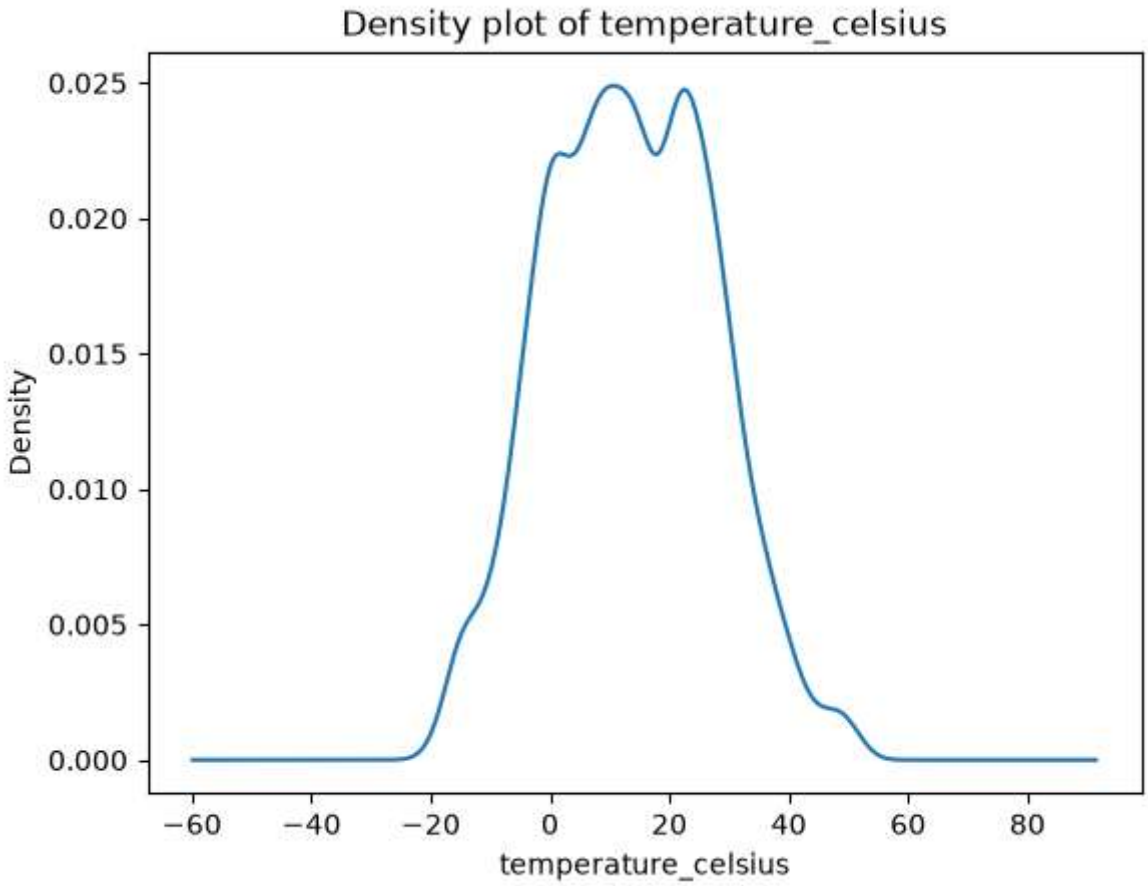

Histogram of humidity_percent
Frequency
0
200
400
600
800
1000
1200
20
30
40
50
60
70
80
90
humidity_percent

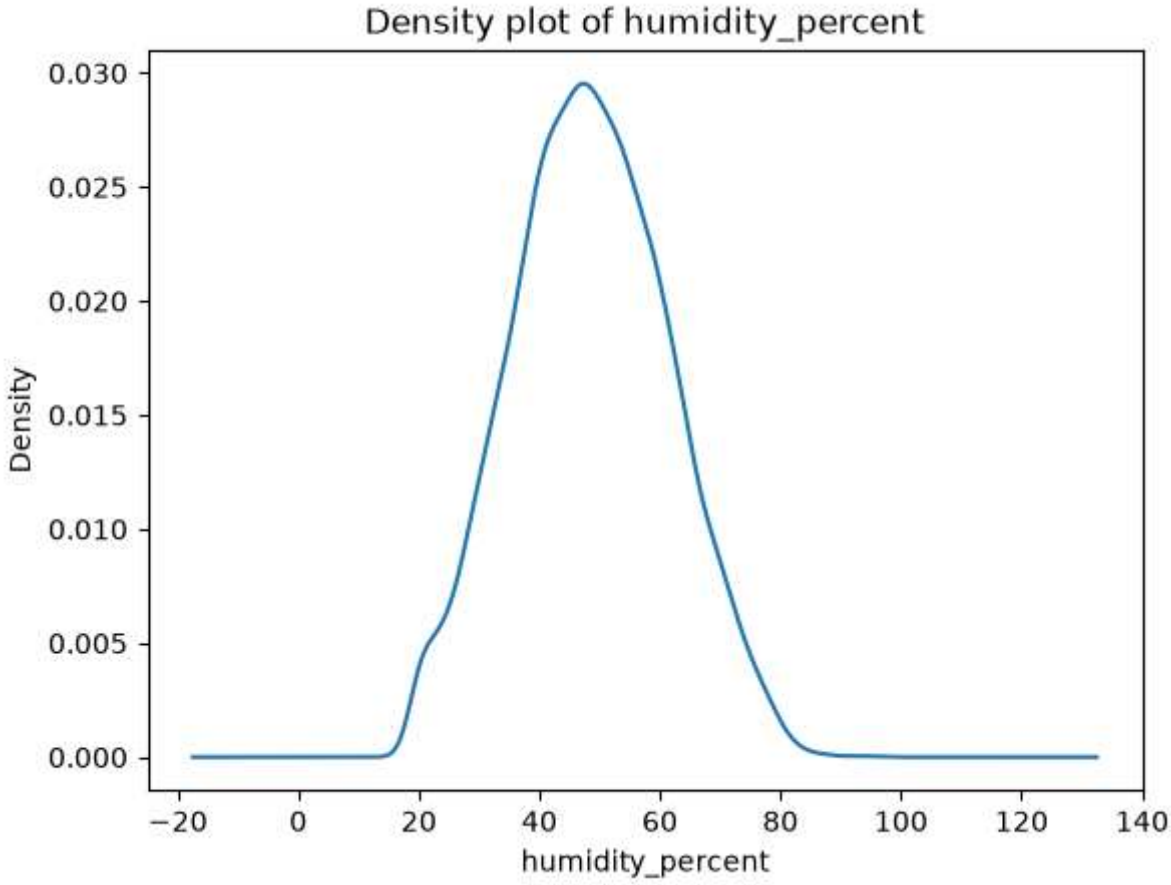


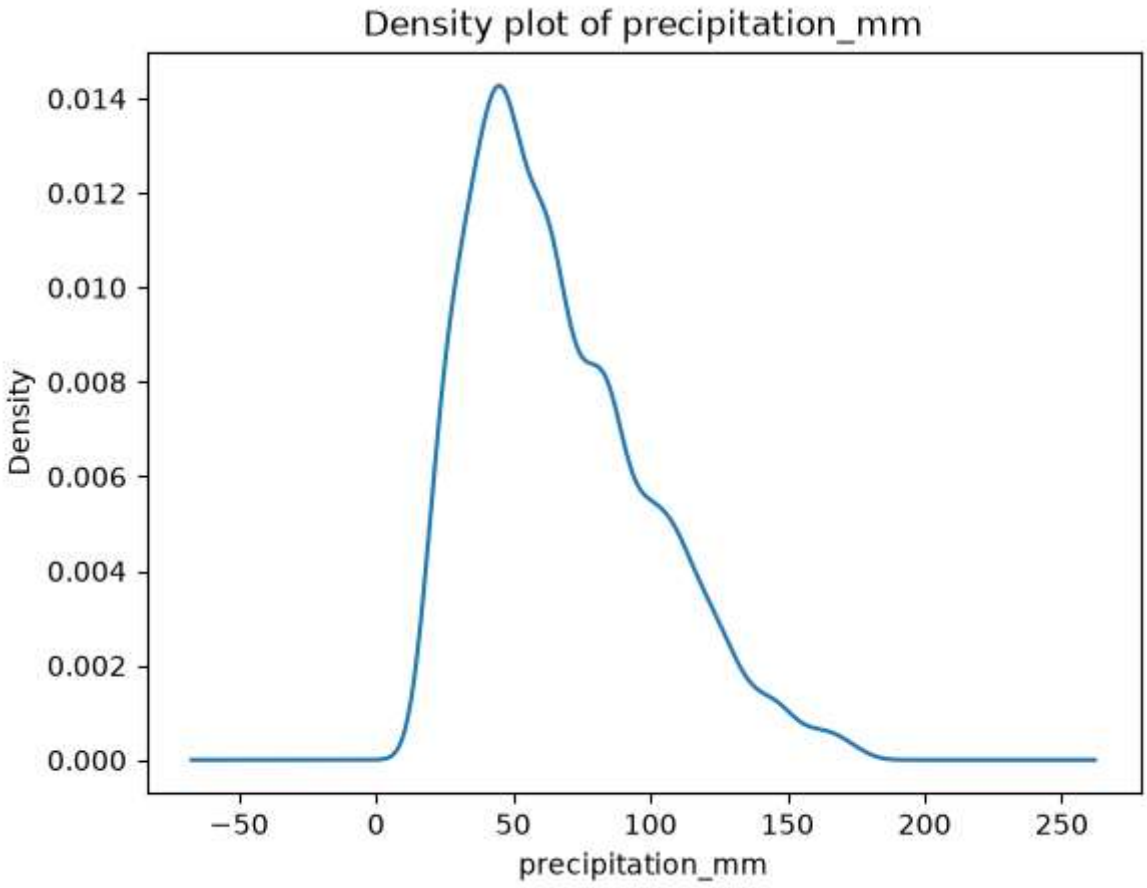
Observation: Describe where values are concentrated and whether each histogram is roughly symmetric, skewed, or has multiple peaks.

5. Generate density plots

In [5]:
```
for column in attributes:
    data = df[column].dropna()
    plt.figure()
    data.plot(kind="density")
    plt.xlabel(column)
    plt.title("Density plot of " + column)
    plt.show()
```


Density plot of temperature_celsius
Density
0.025
0.020
0.015
0.010
0.005
0.000
−60
−40
−20
0
20
40
60
80
temperature_celsius


Density plot of humidity_percent
Density
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−20
0
20
40
60
80
100
120
140
humidity_percent



Observation: A density plot is a smoothed view of the distribution. Peaks indicate ranges where observations are more concentrated.

6. Calculate skewness and kurtosis

In [6]:
```
for column in attributes:
    data = df[column].dropna()
    print("\\n", column)
    print("Skewness:", data.skew())
    print("Excess kurtosis:", data.kurt())
```

```
\n temperature_celsius
Skewness: 0.0848968572253332
Excess kurtosis: -0.4879846415895909
\n humidity_percent
Skewness: 0.07361439430890099
Excess kurtosis: -0.39810893319546875
\n precipitation_mm
Skewness: 0.789652742167261
Excess kurtosis: 0.10599205731627462
```

Earlier calculated observations:

- Temperature: skewness ≈ 0.085, excess kurtosis ≈ −0.488 — nearly symmetric, slight right skew, lighter tails.

- Humidity: skewness ≈ 0.074, excess kurtosis ≈ −0.398 — nearly symmetric, slight right skew, lighter tails.
- Precipitation: skewness ≈ 0.790, excess kurtosis ≈ 0.105 — right-skewed, with excess kurtosis close to zero.

These are the previously calculated moment-based values. Pandas may report slightly different values because it uses adjusted sample formulas.

7. Identify potential outliers using the IQR rule

In [7]:
```
for column in attributes:
    data = df[column].dropna()
    q1 = data.quantile(0.25)
    q3 = data.quantile(0.75)
    iqr = q3 - q1
    lower = q1 - 1.5 * iqr
    upper = q3 + 1.5 * iqr
    outliers = data[(data < lower) | (data > upper)]

    print("\\n", column)
    print("Q1:", q1)
    print("Q3:", q3)
    print("IQR:", iqr)
    print("Lower limit:", lower)
    print("Upper limit:", upper)
    print("Potential outlier count:", len(outliers))
```

```
\n temperature_celsius
Q1: 2.81
Q3: 23.85
IQR: 21.040000000000003
Lower limit: -28.750000000000004
Upper limit: 55.410000000000004
Potential outlier count: 0
\n humidity_percent
Q1: 39.4
Q3: 57.5
IQR: 18.1
Lower limit: 12.249999999999996
Upper limit: 84.65
Potential outlier count: 8
\n precipitation_mm
Q1: 41.7
Q3: 86.92500000000001
IQR: 45.22500000000001
Lower limit: -26.137500000000003
Upper limit: 154.76250000000002
Potential outlier count: 144
```

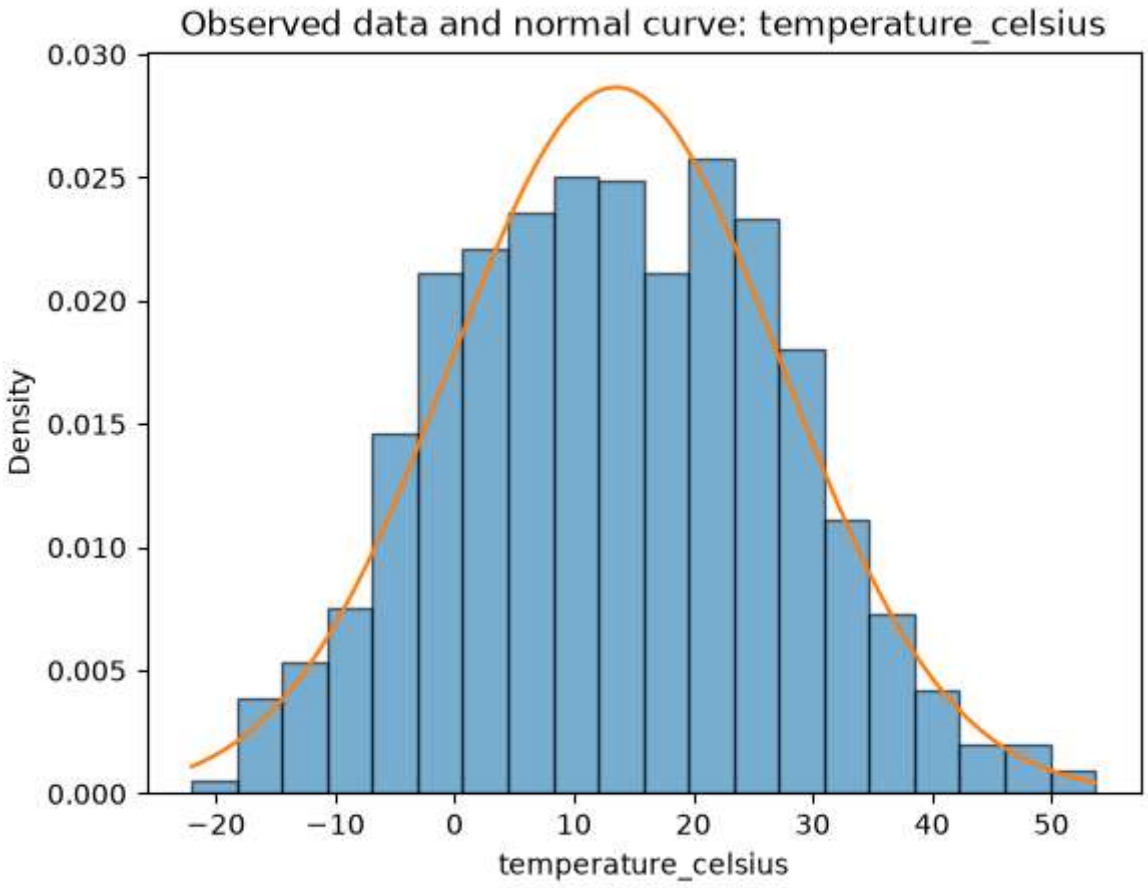
Observation: Values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR are flagged as potential outliers. Your earlier temperature result was Q1 = 2.81, Q3 = 23.85, limits −28.75 to 55.41, and zero potential outliers. Note the humidity and precipitation counts printed by the code.

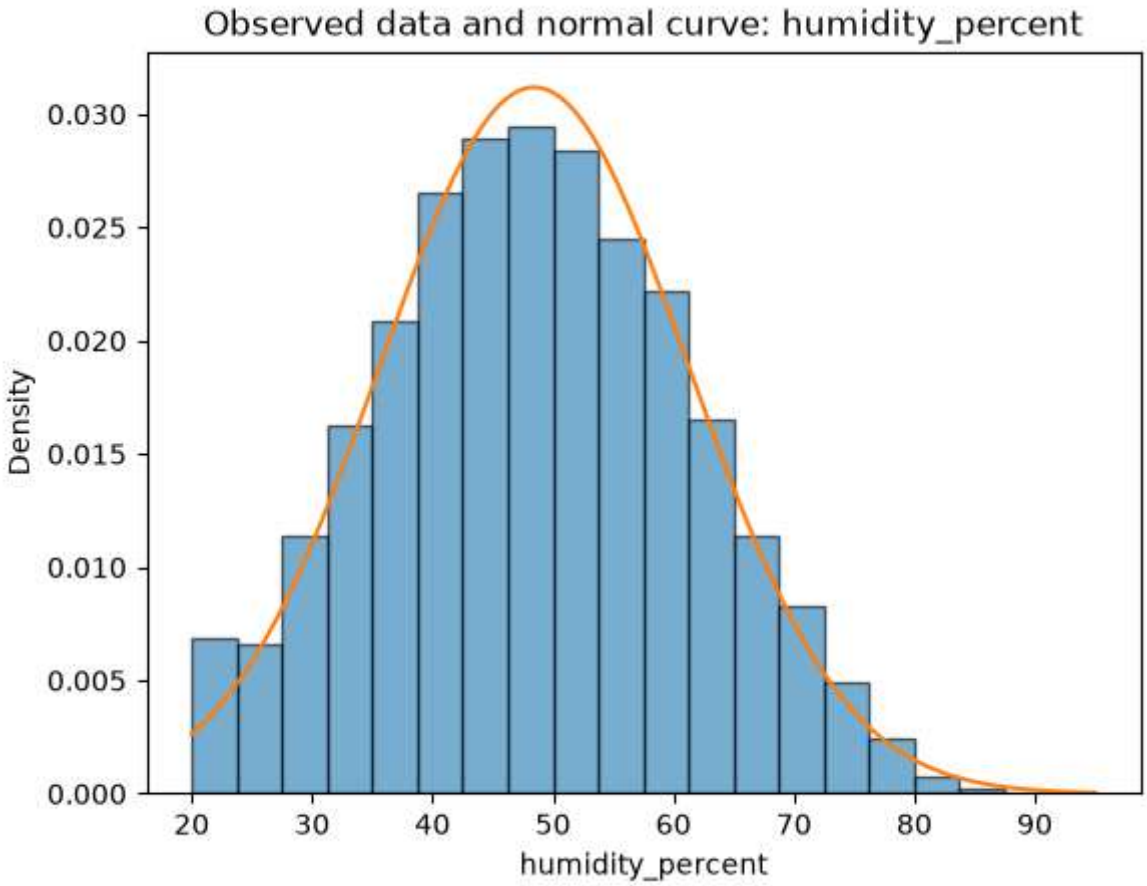
8. Compare observed data with a theoretical distribution

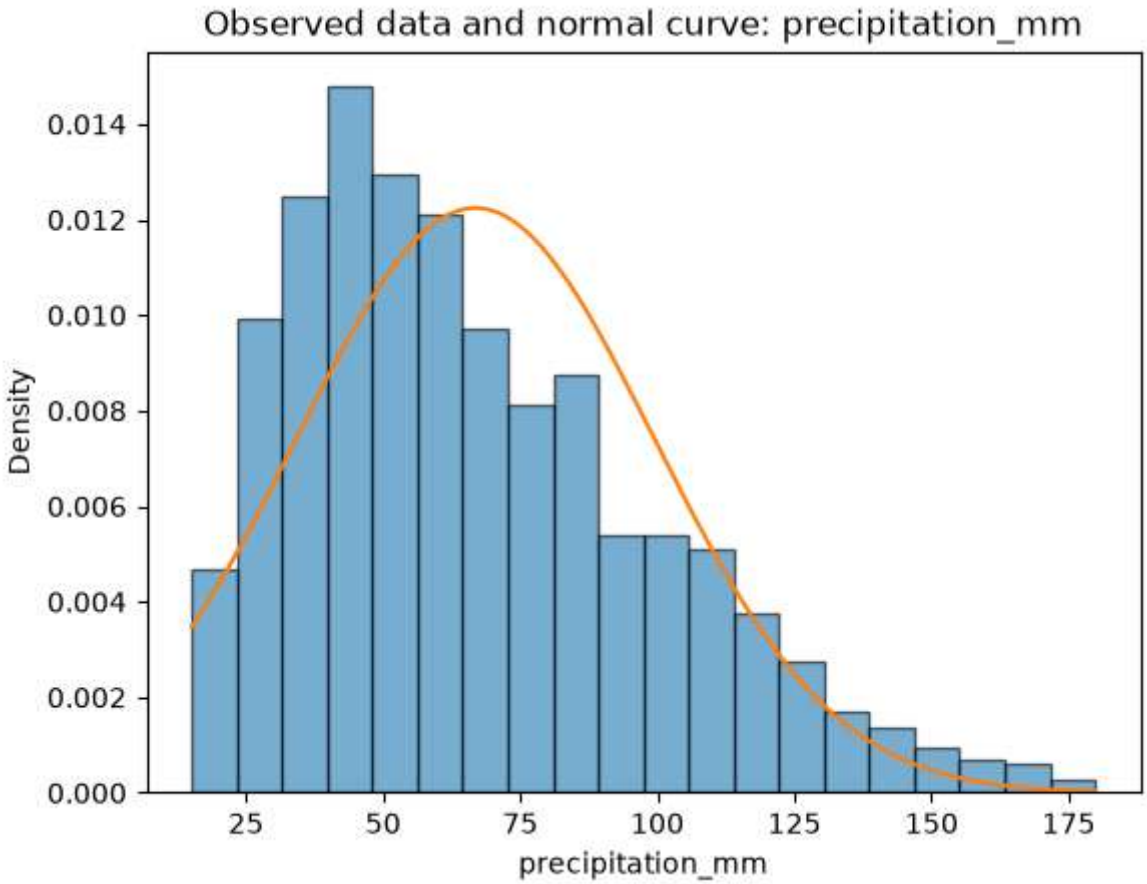
Overlay a normal distribution curve on each histogram. The curve uses the variable's mean and standard deviation.

In [8]:
```
for column in attributes:
    data = df[column].dropna()
    mean = data.mean()
    sd = data.std()

    plt.hist(data, bins=20, density=True, alpha=0.6, edgecolor="black")
    x = np.linspace(data.min(), data.max(), 200)
    curve = (1 / (sd * np.sqrt(2 * np.pi))) * np.exp(-0.5 * ((x - mean) / sd) ** 2)
    plt.plot(x, curve)
    plt.xlabel(column)
    plt.ylabel("Density")
    plt.title("Observed data and normal curve: " + column)
    plt.show()
```




Observed data and normal curve: humidity_percent
Density
0.030
0.025
0.020
0.015
0.010
0.005
0.000
20
30
40
50
60
70
80
90
humidity_percent


Observed data and normal curve: precipitation_mm
Density
0.014
0.012
0.010
0.008
0.006
0.004
0.002
0.000
25
50
75
100
125
150
175
precipitation_mm

Observation: If the histogram bars roughly follow the bell-shaped curve, a normal distribution may be a reasonable approximation. A mismatch suggests that the normal model may not describe the data well. Inspect each graph, especially precipitation, for visible skew.

9. Interpret the distribution of each selected variable

Temperature: Earlier skewness was close to zero, so temperature is nearly symmetric with a slight right skew. Negative excess kurtosis suggests lighter tails than a normal distribution. The IQR check found zero potential outliers.

Humidity: Earlier skewness was close to zero, so humidity is nearly symmetric with a slight right skew. Negative excess kurtosis suggests lighter tails. Use the histogram and IQR output to describe its spread and outlier count.

Precipitation: Earlier skewness was positive (about 0.790), indicating a right-skewed distribution with some higher values extending the right tail. Its excess kurtosis was close to zero. Use the histogram and IQR output to describe its actual shape and outlier count.

10. Discuss how distribution characteristics may affect robotic data analysis

Discussion: Distribution shape, spread, and unusual values affect how sensor data is summarized and modeled. Skewed data can make the mean less representative of a typical reading, so the median may also be useful. A large standard deviation indicates greater variability. Potential outliers should be checked because they may reflect measurement noise, unusual conditions, or data-entry problems; they should not be removed automatically. Comparing observed data with a theoretical curve helps assess whether a statistical model is a reasonable fit.

For this report, these ideas are demonstrated using urban climate variables rather than robotic sensor variables.

Overall conclusion

The experiment used summary statistics, histograms, density plots, skewness, kurtosis, IQR outlier detection, and a normal-curve comparison to describe urban climate data. Earlier calculations showed temperature and humidity are nearly symmetric, while precipitation is more right-skewed. These methods help summarize data, flag unusual observations, and assess possible statistical models.